

ICT
IN CIRCUIT TEST

Abstract :

- In Circuit Testing (ICT) is one of the important assembly processes in manufacturing assembly line and considered to be as manufacturing verification tool in the electronic manufacturing industry. It tests the individual components; Component's inter connections with PCB and to some extent product functional test.

- Though Semiconductor /component manufacturers assures about accuracy and quality of their product , and even though modern technology assembly equipment's like pick and place having additional features –LCR function etc. In circuit testing (ICT) is having its own stand and importance for the functional verification of circuits and individual components.

- For example smart phones. Present era of phones having extra futures like Radio and Camera. It does not mean that dedicated cameras and music systems faded away from the technology. Professional activities still required dedicated equipment. Hence ICT is most important for Circuit and component checks as the substrate, passive and active components pass through various mechanical and thermal stress during the assembly process.

- ICT usually found in typical manufacturing line at the end either in line or off line and it could be together with the Functional test. The main objective is to filter the Manufacturing defects which occur during the process. Hence ICT works as a output of various measuring equipment. Accuracy of measured results classify the performance of the ICT, but outcome of results comply with product specifications is purely depends product circuit design and test points and nodes provided in the PCB circuit by the designer for functional test.

INTRODUCTION ;

ICT is a powerful tool for printed circuit board test. The purpose is to find both component and manufacturing defects before assembly completed. ICT used for both BBT (Bare board test) and after assembly. It is possible to undertake very comprehensive form of printed circuit board test, ensuring circuit has been manufactured correctly and has a very high chance of performing to its specifications.



- ICT carries BBT for PCBs to find shorts, open , trace cross talk and in assembly phase - measures the values of components like Resistors, Capacitors, inductors, Diode or Transistor, Zener diodes, DC voltage, Short or Open measurement , Transistor with built in resistor, FET and Opto couplers .

- Capability of measurement range for each mentioned components differ from ICT manufacturer to manufacturer and model to Model.

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- ICT has two parts. Test performed before power is applied and test performed after power is applied

- Power Off Tests -

  Power off tests are intend to find common problems –Short between trances or component leads , Open circuits where electrical continuity exists, Component values , presence or absence of passive components, test for jumpers and switches and their location and settings and limited testing presence and absence for Active analog components .

  - Powers off tests are performed with separate option generally known as either In Circuit analyzer (ICA) or Manufacturing defect analyzer (MDA).

    - Digital Simulation - Program - Analog simulation

- Power On Tests –

  Power on tests are intend to find common problems in addition to Power off tests listed above – Miss oriented and wrong for analog or digital components, Capacitor and inductance values , Transistor beta value, “stuck” process step and digital component timing.

  Hence ICT generates tests for both Active and Passive components.

- Beyond the basics ICT , ICT can also perform programming –
  - Flash / ISP
  - Boundary scan
  - ICT

  (usually Telecom and Automotive Industries)

- Typical IC with boundary scan –The device can be controlled and Tested through TDI (Test Data In) , TC (Test Clock), TMS(test Mode Select) and TDO (Test Data Out)

In- circuit Test - Over View And Working Principle :

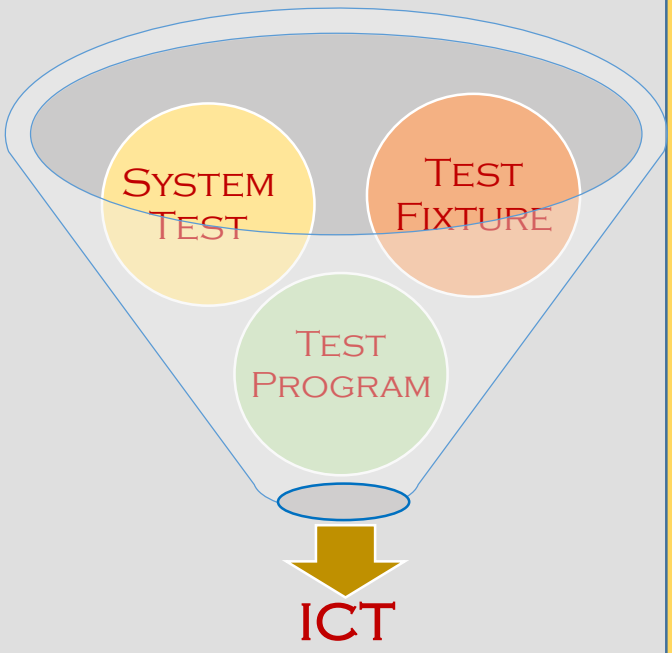
In circuit testing works with an approach of divide and conquer. First the tester checks loaded assembled Board i.e device under test, for unwanted shorts and open. Then it isolates and test each separate component on the Board one at a time .

If all these tests are successful then the Board considered being a 'GOOD' Board. IF one or more tests failed then the Board considered being a BAD board and ICT provides failure details i.e. cause of failure.

Since each type of component tested separately and one at a time, Pin pointing the cause of failure very easy.

In Circuit Test Over View –

Typical ICT tester comprised of Three level components . System Test, Fixture and software.



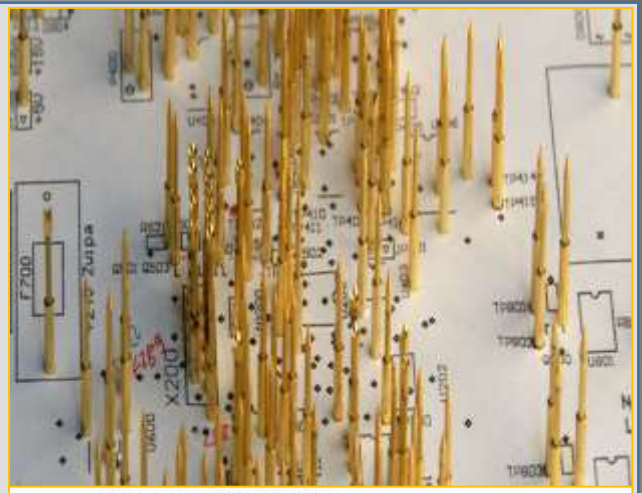
- System Test – performs analog simulation /measurement of subsystem
- Test Program - – Software is written for each board type that can be tested. It instructs the test system what tests to perform between what points and the details of PASS and FAIL Criteria.

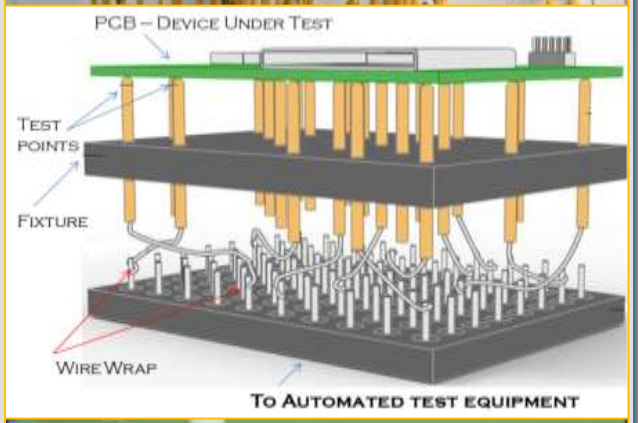
- **FIXTURE** – to access circuit nodes on PC board, the tester uses special test fixture which also called Bed of Nails.

  The whole purpose of bed of nails is to connect the system test instruments to the circuit nodes.

  The nails on this fixture are small spring loaded probes that touches PCB test points, nodes, tracks and component leads on board during test.

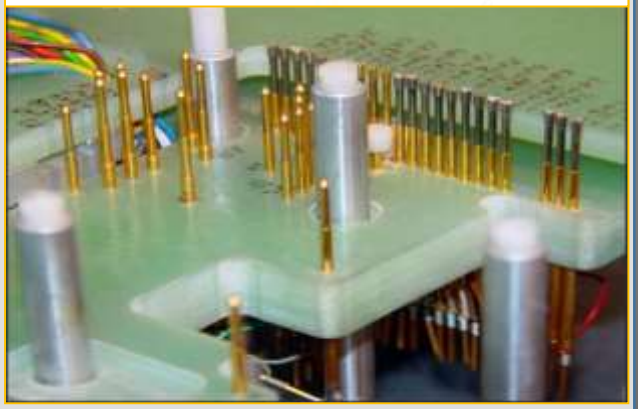
  These fixtures are unique, not common and designed especially for that particular board which needs to be tested in volumes. These fixtures act as an interface between board and circuit tester.





- **IN CIRCUIT TESTER** – The In circuit Test system consists of Matrix Of drivers and sensors that are used to set up and perform the measurements.

  Also consists of Controller and Interface that allow a verity of board fixtures to be placed on the tester which incorporates electrical connection using ZIF (Zero insertion Force) connectors. Other elements like air, vacuum, high and low power source



Basic Understanding the Concept of How ICT works and Measures

- ICT must have access to all circuit nodes on board and the tester must be able to connect test instruments to each pin of the component.
- Also it requires special isolation techniques to prevent component under test from being effected by other component.

  For better understanding, let us analyze the following Figures with considering an example of measuring Resistance of a resistor , capacitance of capacitor or inductance of Inductor .

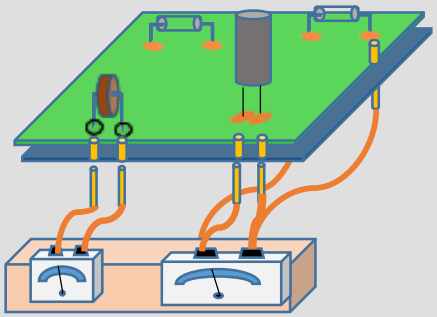


Figure 1

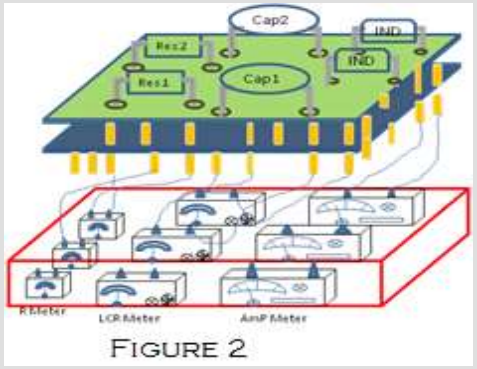


Figure 2

- Figure 1- Test PCB or device under test is placed on the Fixture (bed of Nails). Test probes connected Resistor terminals to the instrument Ohm meter and also probes connected from LCR meter to Capacitor nodes and inductance.

- Figure 2 - Here in this case we have more than multiple resistors , capacitors and inductors are in device under test. ICT thumb rule is ICT measures the component values one at a time . Hence we need multiple measuring instruments of each type in ICT system. Measurements taken and the results stored in memory and report generated by Fig 2.

- Now the question is how we can measure hundreds of components on one particular assembled PCB where it has thousands of nodes , and required various types of measurements by maintaining the volumes and cycle time ?

As per basic concept, ICT system requires many varieties of instruments in quite large number in Quantity to measure all the component values which is QUITE IMPOSSIBLE. Now let us consider figure 3.

FIGURE 3 - In this case , ICT system having only one Ohm meter, LCR meter as shown in Fig1 but Device under test having multiple resistors and capacitor or inductors(i.e various type of analog and digital instruments with only one qty).

Here we introduced 4 -6 pairs of conductors (can be called as Channels BUS), a pair of instrument terminals conned with a pair of trace lines and in the same way LCR and other Instrument. i.e. A &B lines connected with Ohm meter and all the test points of Resistor components are connected to Line A& B . In the same way Lines C& D are connected to LCR instrument and all capacitors terminals to C &D.

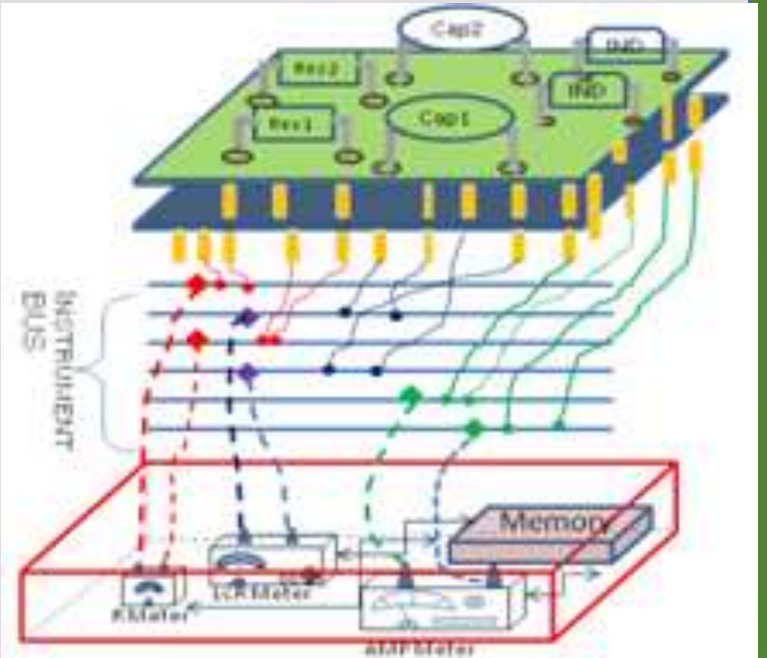


FIGURE 3

- Let's have a look what Analog and Digital test instruments that are used by the ICT and how these instruments are routed to the circuit nodes on the Board.
- To perform analog testing, the tester uses – DC current source, DC voltage source, DC Voltmeter, DC Ammeter, AC Impedance hl Measurement where as Drivers and Sensors used as digital instruments.
- Now let us understand the techniques how the tester connects any one of these instrument to any one of hundreds of circuit nodes on the Board?

  It is all done by the computer controlled and directed by the test program uniquely developed by the tester for that particular board.

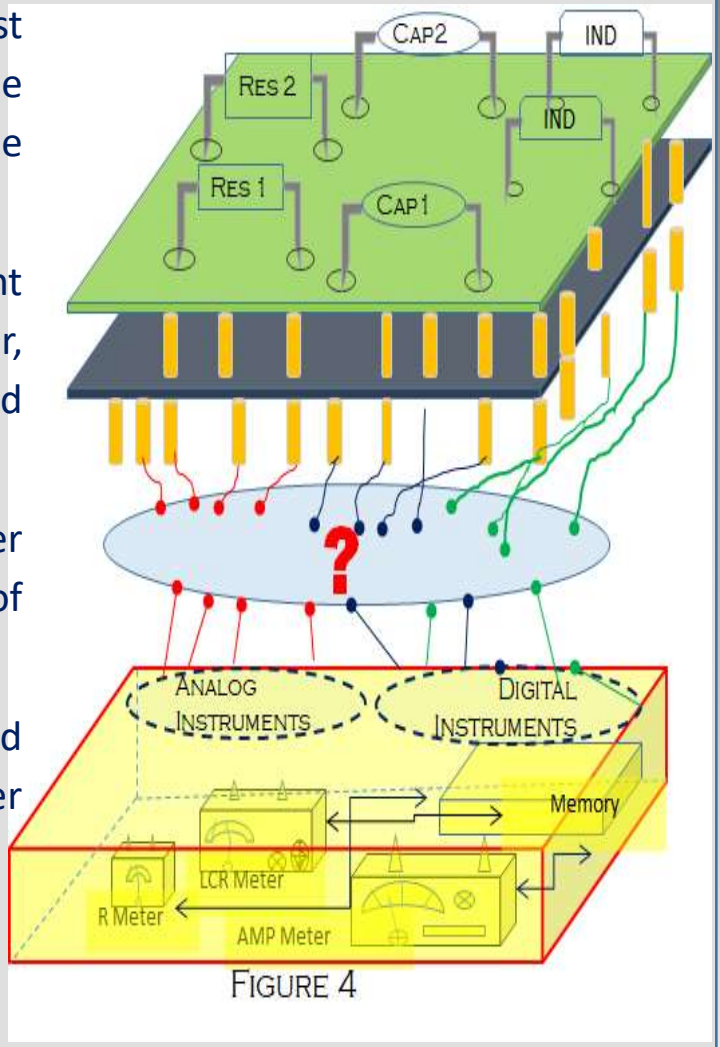


FIGURE 4

Connecting analog test instruments to nodes –

- There are two sets of relays , one set called scanner and the Other set called Instrument Mux. Scanner connects any test nails to any of the Instrument Bus channels. The other set of relays called as Instrument multiplexer or MUX connects these same Instrument Bus Channels to any of analog instrument.
- So the actual connection between test instrument and test nails are made by program controllable relays i.e., switches that are opened and closed by program instructions These relays Route both Test instrument and Test Nails to a common set of channels Called as Instrument BUS.

❖ **Multiplexing** - Today's PCBAs very complicated and node count can easily rise over a thousand and may reach several thousand on some.

To have dedicated pins on the tester for each node can be very costly as each one requires its own driver sensor. To reduce this manufacturers introduce a system known as multiplexing.

Here a particular node may be placed through a switching matrix so that it can address more than one node. The number of nodes that are addressed by each tester primary node is known as the multiplex ratio.

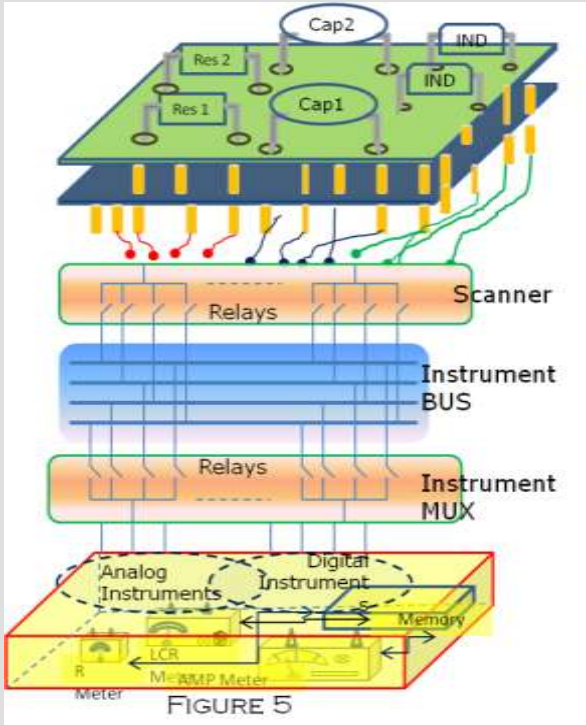
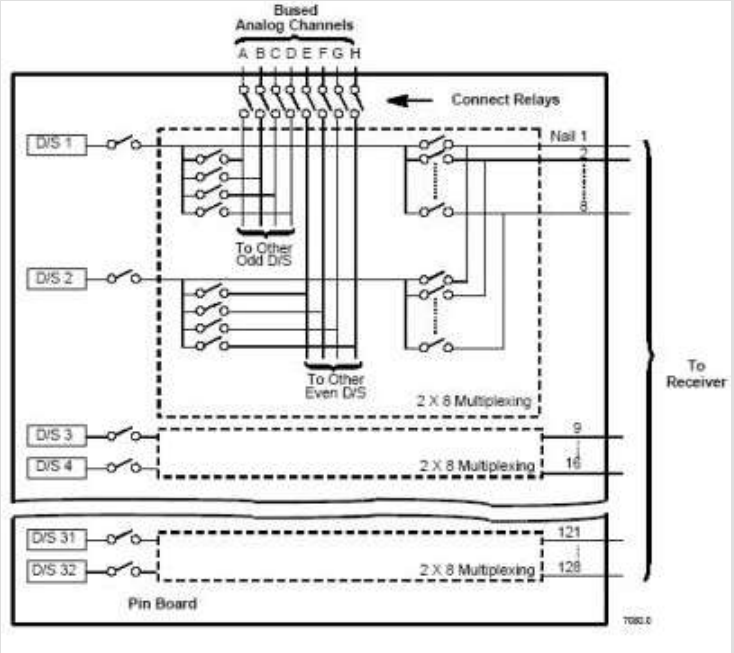


FIGURE 5

What is Multiplexing –

- A technique for sharing (or pooling) tester instruments among a large number of test points
- Multiplexing increases the number of test points that have access to ICT instruments
- Multiplexing places rules on how test instruments can be accessed during testing.



❖ Driver-sensors for ICT –

- Driver-sensors are the active circuits that are used for making the measurements. Normally drivers and sensors are always present in pairs in an in-circuit test system.

- Drivers supply a voltage or current to enable a node in the circuit to be driven to a particular or the required state despite the condition of the surrounding circuitry. Typically they may need to force the output of a digital IC to a given state despite the natural output state of the device. To achieve this, the output impedance of the driver must be very low.

❖ Guarding -

- The key to the success of ICT is a technique known as guarding. It is very easy to measure the value of a component when it is not in a circuit.

- For example a resistor value can be measured by simply placing an ohmmeter across it. However when the component is in a circuit, the situation is somewhat different. Here it is most likely that there are other paths around the component that will alter the value that is measured.

- To overcome this problem and gain a far more accurate indication of the value of the component a technique known as guarding is used. Here the nodes around the component under test are earthed and in this way any leakage paths are removed and more accurate measurements made.

Understanding Basic Analog Measurements:

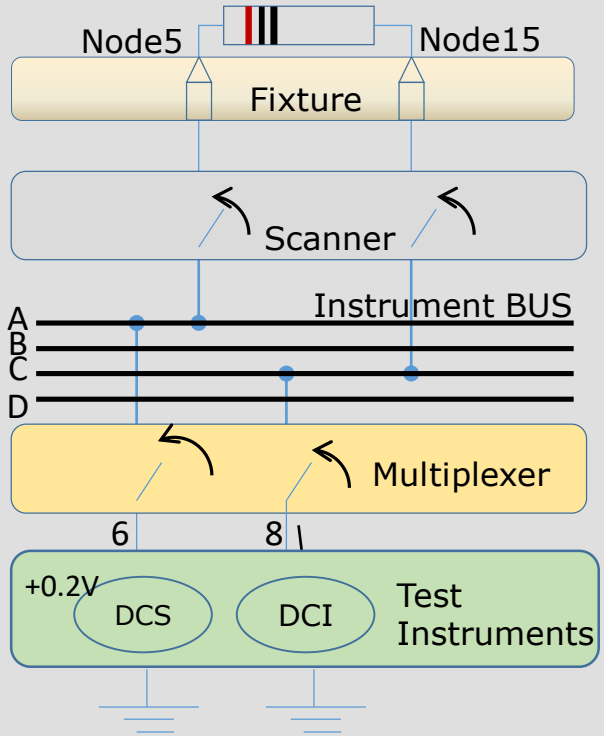
- Let us consider analog test example for clear understanding on Basic Knowledge. Let us say PC board has a resistor connected between nails 5 and 15. And you want to apply **0.2 V** across the resistor, measure the resulting current and store **the** value **in** a memory location.

Set SCAN AT (ChA =5 : CHC = 15):

The functional sequence can be
SET MUX **AT** (CHA=6: CHR=8):
SET **SCAN** AT (CHA=5**:** CHR=15);
SET DCS V-0.2:
MEASURE DCI RESULT AND STORE INTO MEMORY.

Set MUX AT (ChA -6 : CHC – 8):

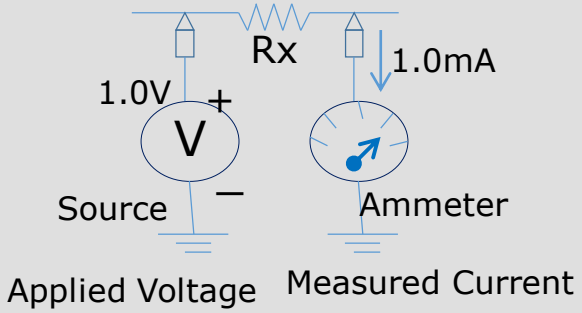
Set DCS +0.2



Concepts Of Isolation For Measurement :- Basic understanding

Case1 – two terminal measurements.

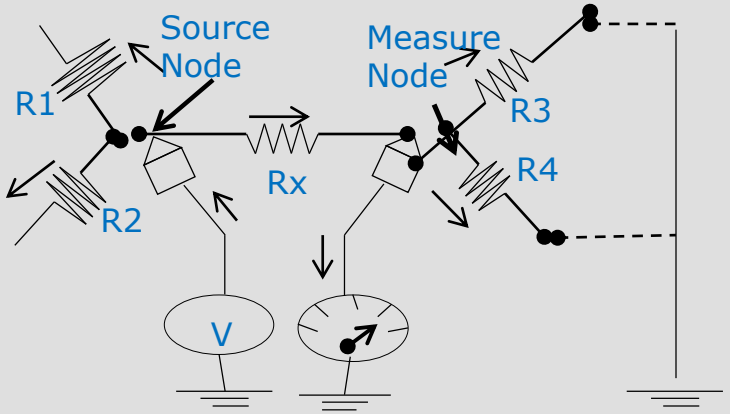
If applied voltage were 1V (source Voltage) and the measured current 1mA , then as per Ohms law , R =V/I i.e., Rx = 1.0/0.001 =1000Ohms.

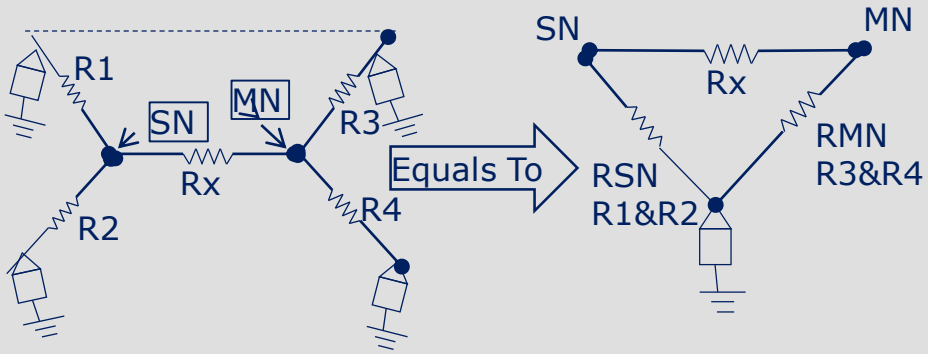


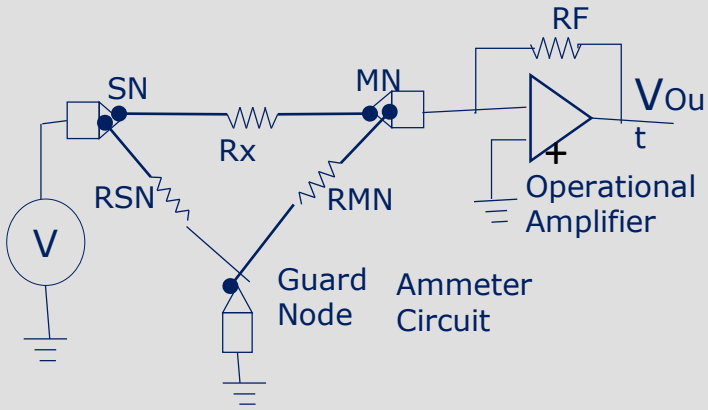
- Case 2 – If the same resistor connected with circuit, chances some other components connected in parallel.

  Let us consider R1,R2,R3 and R4 are in shunt path. Current distributes through R3 , R4 and ammeter. Hence Rx Value cannot be measured.

  So we need to use Guard against unwanted shunt currents .this method stops unwanted shunt paths to get accurate value of Rx.







Guarding is accomplished by –

1. Temporarily connect all shunt paths around Rx(component under test) to ground (guard voltage)
2. Use operational amplifier in a ammeter circuit .This is called three terminal measurement

How does OP Amp help in getting Value of Rx ?

- Because of feedback resistor (Rf), the negative input to the amplifier is forced to be very nearly equals to positive input. Since positive input is tied to ground , the negative input is also very close to ground (virtual ground).

- In the ideal case, the measured Node(MN)is at 0Volt(virtual ground) due to operational amplifier and guard node is also ZERO volts because it is connected to ground by the tester. Since there is voltage across measure and guard nodes, obviously No current. In this way we have isolated Rx from shunt paths. All the current through Rx now flows through ammeter.

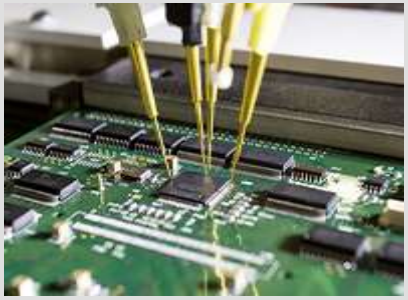
- The tester knows the value of Rf feedback resistor and can measure the Op- amp Vout. Therefore determining the current flow into the ammeter circuit , Tester can calculate Rx value .

Types Of In Circuit Testers:



- The main types of ICT machines that are available and the requirement type is depend on the manufacturing or test process used , the volume and Boards that are Used .

1. Standard ICT Machine - Generally called IN Circuit Tester . Widely accepted test machine



2. Flying Probe Tester - This has a simple fixture to hold the board and contact is made via a few probes that can move around the board and make contact as required. These are moved under software control so any board updates can be accommodated with changes to the software program.



3. Manufacturing defect analyzer, MDA - This form of tester offers a basic In-circuit test of resistance, continuity and insulation. It is just used for the detection of manufacturing defects like short circuits across tracks and open circuit connections.

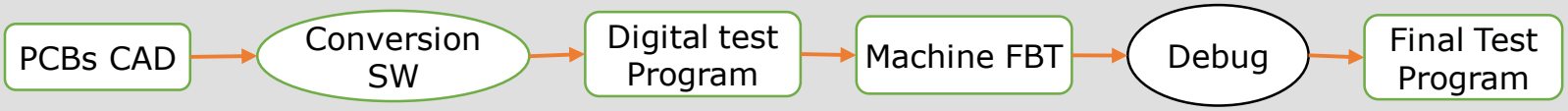
4. Cable form tester: This form of tester is used to test cables. It uses the same basic functions as an MDA, although some form of high voltages may need to be applied occasionally to test for insulation. Its operation is optimized for the testing of cables.

I C T Test sequence

- Discharging Capacitors (Electrolytic capacitors)
- Contact Test (Device under test)
- Short Testing (Test for solder shorts and open)
- Analog Test (Test all analog components for placement and correct Value)
- Test for defective open pins on devices
- Test for capacitor orientation defects
- Power Up Device under Test
- Powered Analog (Test for correct operation of regulators and OPAmps
- Powered Digital(Test operation of digital components and boundary scan devices)
- JTAG Boundary scan tests
- Flash memory , EEPROM and other device programming
- Discharge capacitors (DUT is powered down)

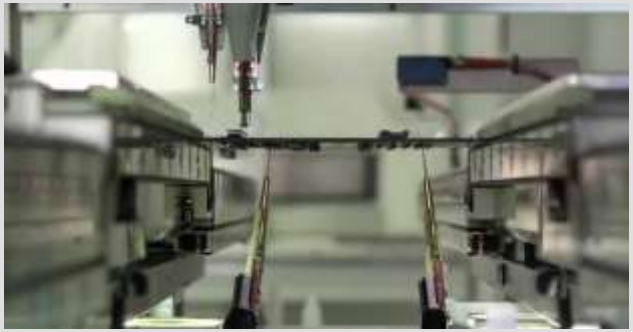
Flying Probe Tester -

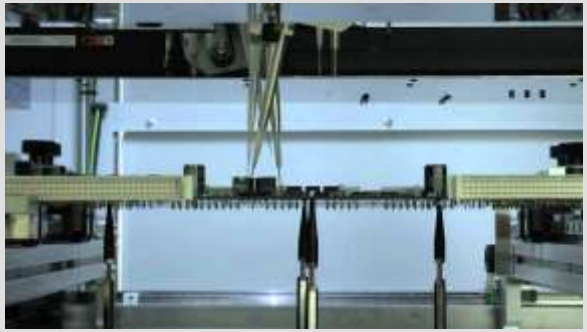
- The system uses a generic board holder, and one or more probes moves across the board accessing individual nodes under software control.
- Flying probe testers now include facilities such as on-board memory module programming and boundary scan testing. With these capabilities, they are able to offer the equivalent performance of an advanced in-circuit tester.
- Working principle/ method –



- Overall accuracy ± 1.25µm which helps to reduce the DFT Design For Test
- Machine models comes with different optional configurations –

  1) 4 Tilted Probes 2) 4 tilted + 2Vertical 3) 4 Tilted + 2 Vertical +2 sensors
- LED Color test can also be possible . Measures colors and its Brightness
- Some Models also checks for AOI 2dimentional





Flying Probe Tester - Advantages

Advantages –

- ✓ No special fixture required - Fixture less , Low volume High Mix , no need Of Test points ,Bed Of Nails and Low running cost . The probes move according under software control to make contact with the required nodes.

- ✓ Changes can be made easily - As the probes move under software control, any changes to pad positions or components can be made by purely changing the software. It is not necessary to make any mechanical changes to a fixture as in the case of a "bed of nails" fixture. Flexible response to design change. Fast Test arrangement Fast Model change .

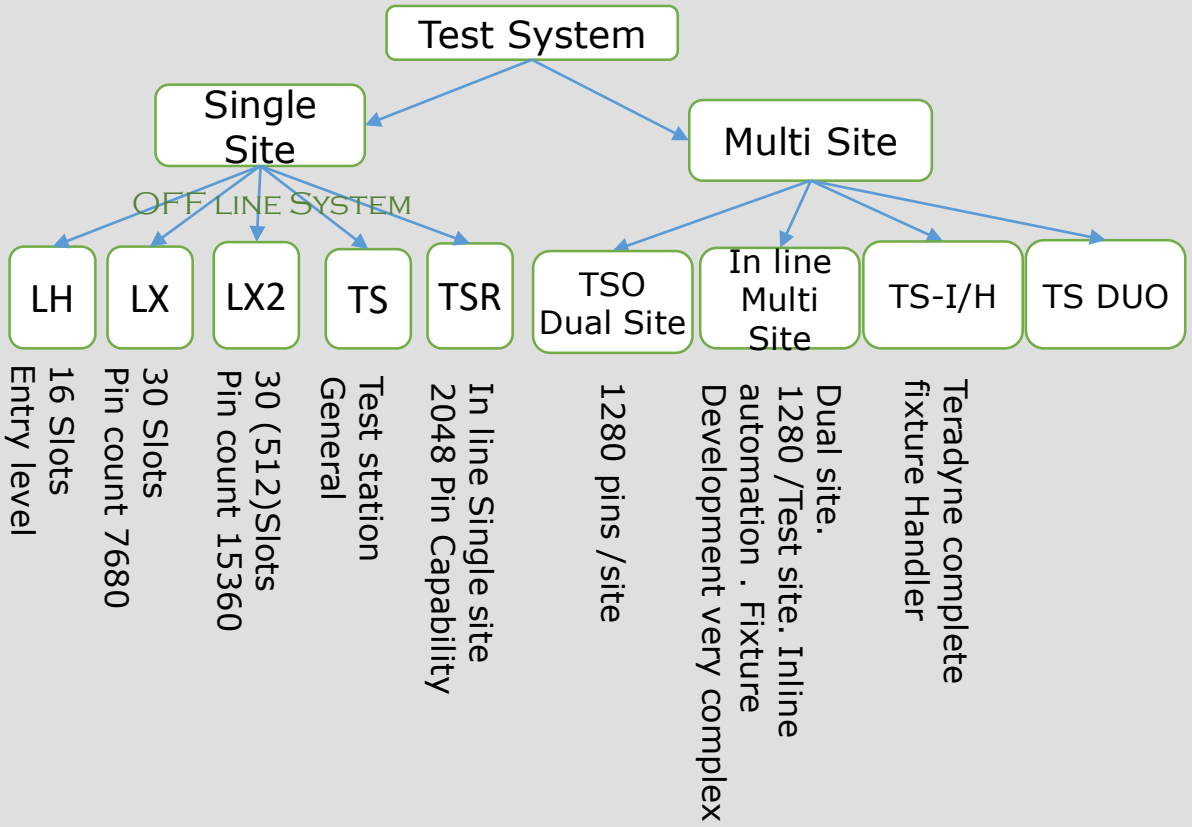
- ✓ Test development time reduced - Software for a flying probe tester can be developed relatively quickly from the PCB design files. The big saving is that no mechanical fixture is required and its manufacturing time is not needed. Accordingly development of the test program for the flying probe tester simply requires the PCB files, and ultimately a first off board or boards on which to test the program.

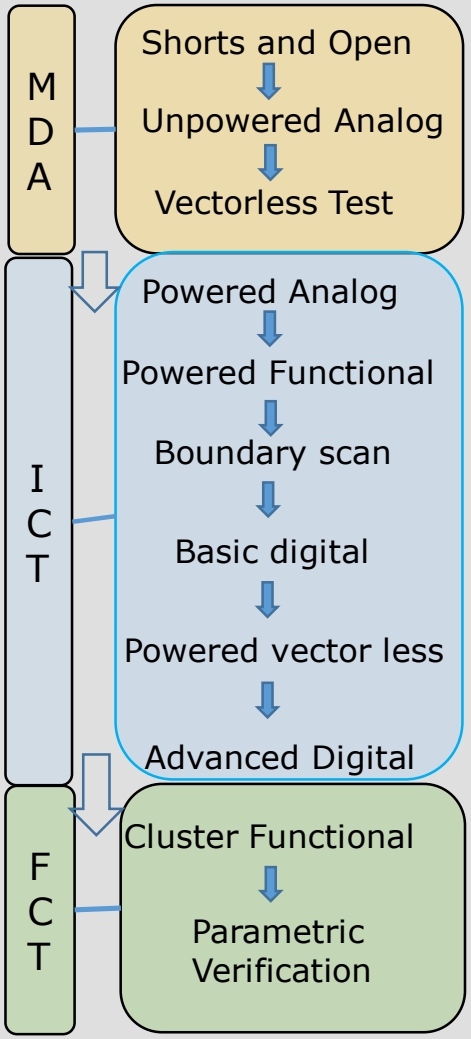
Disadvantages –

- ✓ It may not always possible to make complicated tests - Using early flying probe testers it was not possible to test components beyond passive components or diodes. To achieve higher levels of fault detection technologies such as boundary scan and the use of on-board memory enable more complicated tests to be undertaken. It is necessary to check the performance of the individual flying probe tester to ensure it can meet the requirements.

Know about Product ICT –Teradyne

- Inetest representing the segment of –Product test system
- Provides safety accurate testing and Low voltage technology
- High Fault coverage 1 Pin count and test through pins
- Designed with scalable system architecture
  - ✓ Compatible software
  - ✓ Compatible Hardware
  - ✓ Compatible Programs



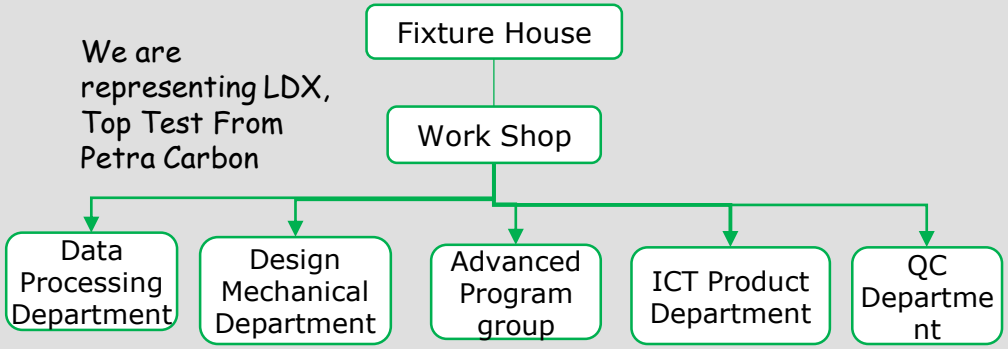


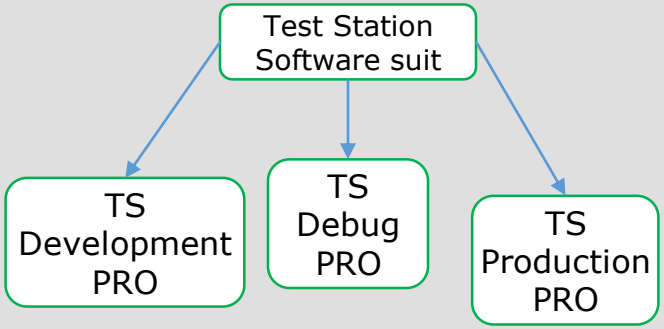
Discussion Points -Teradyne

With the addition of these pin-cards, Teradyne's TestStation now offers a choice of options that include *MUX* (multiplexed) or pure-pin, analog only or hybrid, and low pin-count or high pin-count configurations.



The UltraPin II 121 has 128 non-multiplexed digital/analog pins per board. It supports a maximum of 2048 pins when configured in a *TestStation LH*, or 3840 pins when installed on the company's *LX* model ICT.





Debug Pro also has a *Guard Editor* window that displays test instrument connections and relevant in-circuit interconnections associated with *DUT*s (devices under test). You can add and remove test guard points using the mouse. Debug pro also includes a *Charting Tool* that displays measurement results using run or histogram charts

Debug Pro features an intuitive *GUI* (graphical user interface) and comprehensive tools with simplified command menus, structures, and operational displays.

A key element of Debug Pro is a *Label Navigator* that graphically displays each of the program's tests

Debug Pro also includes a *Statement Editor* that displays test-specified parameters available for user-selected tests

- **Pre Requisites from the Customer**
  a) CAD / BOM / Schematic / Gerber / Golden Board
  b) Panel /single
  c) ISP
  d) LED
  e) BSCAN
  f) Stain gauge
  g) Functional Test
  h) Details of the systems Customer having

Thank You

❖ Cluster Testing

- Groups of two or more components are tested as a single entity to verify proper operation
- Allows testing of functions like RC Networks and filter circuits without access to all pins in the cluster

❖ Combined BSCAN + Framescan [7]

- Combines BSCAN and capacitive opens test techniques. BSCAN devices provide digital stimulus, probe sensor plate in fixture detects stimulus signal.
- Identifies open pins on connectors, sockets and IC devices that do not have physical test access

❖ Boundary Scan / Embedded Test

- Many boards today have components with built-in testability features like boundary scan that can be activated during in-circuit testing to run board self tests or provide virtual access to signals on the board when the ICT system does not have physical test access.
- Detects faults on nets that do not have physical test access. Uses ICT resources and board testability features in concert for maximum test effectiveness. Consolidates tests on single manufacturing test solution